Rationality, common knowledge and dominance

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Game Theory and Mechanism Design - Lecture 3

(This lecture heavily relies on Tadelis (2013) Game Theory – An Introduction – chap. 4)

Learning objectives

At the end of this lecture (and related tutorial(s)), a student should be able to:

- Describe the concept of **dominance** in normal form games
- Link it to **rationality** and **common knowledge**
- Apply the **iterated elimination of strictly dominated strategies**
- Understand and discuss the concept of **iterated elimination equilibrium**

Outline

1. What is the solution?
2. (strict) Dominance and dominated strategies
3. (weak) Dominance and dominated strategies
4. Iterative elimination

1. **What is the solution?**

Prisoner dilemma

Remember that the solution is the strategy profile which we believe would be applied in practice...
(the sets of pay-offs are the *outcome* instead)

- 2 individuals are arrested for a crime (P1 /P2)
- They are both interrogated individually and without possibility of communication between them
- The interrogator tells each of them privately that
  - None of them confess and both get 2 years in jail
  - if confessing for both of them, while the other one does not do so, the one confessing will get a single year in jail, and the other one 10 years
  - if confessing for both of them, and they both do so, they will both only get 4 years in jail
- Write the 2 possible actions C (**C**onfess) and NC (**N**ot **C**onfess)

In **matrix form**:

		P2	
		C	NC
P1	C	$-4, -4$	$-1, -10$
	NC	$-10, -1$	$-2, -2$

What is the solution? and why?

What about this one?

Remember that the solution is the strategy profile which we believe would be applied in practice...
(the related set of pay-offs defines the outcome instead)

Now, consider this normal form game, with the following matrix representation:

		player 2		
		L	C	R
player 1	U	$4, 3$	$5, 1$	$6, 2$
	M	$2, 1$	$8, 4$	$3, 6$
	D	$3, 0$	$9, 6$	$2, 8$

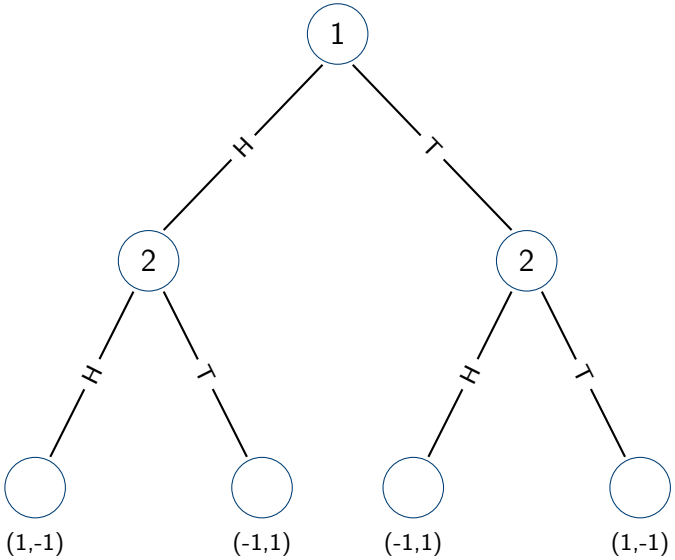
What is the solution? and why?

2. **(strict) Dominance and dominated strategies**

(we only need for the players to be rational)

Dominance (in pure strategies)

We can clearly illustrate this phenomenon with the matching pennies game (with observation)



Matching pennies (with observation):

- 2 players (and their coins)
- player 1 starts by choosing a side for their coin (H or T)
- player 2 observes it and choose a side for their coin (H or T)
- If the 2 players have the same side, the player that played first gets a *pa-yoff* of 1, and the other one of -1
- If the 2 players have different sides, the player that played first gets a *pay-off* of -1, and the other one of 1

If being player 2, what strategies would you eliminate, and why?

Strictly dominated strategies

If being **rational**, they are strategies that a player will simply discard...

Definition (Strictly dominated strategy)

Let s_i, $s_i' \in S_i$ be possible strategies for player i. The strategy s_i' is *strictly dominated* by s_i if, for any combination of the other players' strategies, $s_{-i} \in S'_{-i}$, the pay-off of player i is strictly less if playing s_i' instead of s_i, i.e.,

$$u_i(s_i, s_{-i}) > u(s_i', s_{-i}), \quad \forall s_{-i} \in S_{-i}$$

As a notation, we write $s_i \succ_i s_i'$ to denote that s_i' is strictly dominated by s_i.

Remember notations:

- s_i is a possible strategy for player i, and S_i is the set of possible strategies
- S_{-i} is the set of all possible strategies for all other players (from 1 to n, except for player i)
- $s_{-i} \in S_{-i}$ is one of the possible strategies (for all other players), from this set

Proposition (Rationality and strictly dominated strategies)

If player i is rational, player i will **never** *play a strictly dominated strategy.*

Strictly dominated strategy in the prisoner dilemma

Based on this matrix representation:

		player 2	
		C	NC
player 1	C	$-4, -4$	$-1, -10$
	NC	$-10, -1$	$-2, -2$

It is clear that for both players, **NC is a strictly dominated strategy**... (here, for player 1)

> You can interpret it as: "*Regardless of the strategy chosen by the other players*, it can **never pay off** to choose that strategy"

Strictly dominated strategy in the prisoner dilemma

Based on this matrix representation:

		player 2	
		C	NC
player 1	C	$-4, -4$	$1, -10$
	NC	$-10, -1$	$-2, -2$

It is clear that for both players, **NC is a strictly dominated strategy**... (here, for player 2)

> You can interpret it as: "*Regardless of the strategy chosen by the other players*, it can **never pay off** to choose that strategy"

Another example

This is a marketing campaign game for 2 companies and 3 marketing strategies (low – L, medium – M, and high – H), with the following matrix representation:

		player 2		
		L	M	H
player 1	L	6, 6	2, 8	0, 4
	M	8, 2	4, 4	1, 3
	H	4, 1	3, 1	2, 2

Do you see strictly dominated strategies, for player 1 and for player 2?

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		player 2		
		L	M	H
player 1	L	6, 6	2, 8	0, 4
	M	8, 2	4, 4	1, 3
	H	4, 1	3, 1	2, 2

Do you see strictly dominated strategies, for player 1 and for player 2?

- for player 1, **L** is strictly dominated by **M**
- for player 1, M and H and not strictly dominated (e.g., if player 2 plays M, then player 1 should also play M, and if player 2 plays H, then player 1 should also play H)...
- And, for player 2, there is no strictly dominated strategy

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This is a marketing campaign game for 2 companies and 3 marketing strategies (low – L, medium – M, and high – H), with the following matrix representation:

		player 2		
		L	M	H
player 1	L	$6, 6$	$2, 8$	$0, 4$
	M	$8, 2$	$4, 4$	$1, 3$
	H	$4, 1$	$3, 1$	$2, 2$

Do you see strictly dominated strategies, for player 1 and for player 2?

- for player 1, L is strictly dominated by M
- for player 1, M and H and not strictly dominated (e.g., if player 2 plays M, then player 1 should also play M, and if player 2 plays H, then player 1 should also play H)...
- And, for player 2, there is no strictly dominated strategy

Strictly dominant strategies

Inversely, if being **rational**, they are just strategies that a player will necessarily prefer...

Definition (Strictly dominant strategy)

$s_i \in S_i$ is a *strictly dominant strategy* if all other strategies $s_i' \in S_i$ are strictly dominated by it, i.e.,

$$u_i(s_i, s_{-i}) > u(s_i', s_{-i}), \quad \forall s_i' \in S_i,\ s_i' \neq s_i, \quad \forall s_{-i} \in S_{-i}$$

You can interpret it as: "*Regardless of the strategy chosen by the other players*, this strategy will **always** yield the **maximum payoff**"

Proposition (Rationality and strictly dominant strategies)

If player i is rational, player i will **always** *play a strictly dominant strategy.*

Strict dominant strategy equilibrium

If all players feel they have a strategy they would prefer, regardless of what the others do, it should be possible to predict what the solution (and outcome) of the game is(!)

Definition (Strictly dominant strategy equilibrium)

The strategy profile $s_i^* \in S$, $s_i^* = (s_1^*, \ldots, s_n^*)$ is a *strict dominant strategy equilibrium* if, for all players $i = 1, \ldots, n$, s_i^* is a strictly dominant strategy.

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And, if such an equilibrium exists, could there be several of them?

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And, if such an equilibrium exists, could there be several of them?

Proposition (Uniqueness of the strict dominant strategy equilibrium)

If the game $\Gamma = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$ has a strictly dominant strategy equilibrium s^*, then it is **unique**.

9 (weak) Dominance and dominated strategies

(we only need for the players to be rational)

Weak dominance

Weak dominance only means that we do no have **strict inequalities**

Definition (Weakly dominated strategy)

Let s_i, $s_i' \in S_i$ be possible strategies for player i. The strategy s_i' is *weakly dominated* by s_i if, for any combination of the other players' strategies, $s_{-i} \in S'_{-i}$, the pay-off of player i is less, or equal, if playing s_i' instead of s_i, i.e.,

$$u_i(s_i, s_{-i}) \geq u(s_i', s_{-i}), \quad \forall s_{-i} \in S_{-i}$$

Definition (Weakly dominant strategy)

$s_i \in S_i$ is a *weakly dominant strategy* if all other strategies $s_i' \in S_i$ are weakly dominated by it, i.e.,

$$u_i(s_i, s_{-i}) \geq u(s_i', s_{-i}), \quad \forall s_i' \in S_i,\ s_i' \neq s_i, \quad \forall s_{-i} \in S_{-i}$$

Weak dominance in the prisoner dilemma

Based on this matrix representation:

		player 2	
		C	NC
player 1	C	$-5, -10$	$-2, -10$
	NC	$-5, -2$	$-2, -2$

It is clear that for both players,

- C is a weakly dominant strategy
- NC is a also weakly dominant strategy

So, **what do you think the solution may be?**

Weak dominant strategy equilibrium

If all players feel they have a strategy they feel is at least as good as others, regardless of what the others do, it should also be possible to predict what the outcome of the game is(!)

Definition (Weakly dominant strategy equilibrium)

The strategy profile $s_i^* \in S$, $s_i^D = (s_1^*, \ldots, s_n^*)$ is a *weak dominant strategy equilibrium* if, for all players $i = 1, \ldots, n$, s_i^* is a weakly dominant strategy.

Weak dominant strategy equilibrium

If all players feel they have a strategy they feel is at least as good as others, regardless of what the others do, it should also be possible to predict what the outcome of the game is(!)

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And, if such an equilibrium exists, could there be several of them?

Proposition (Non-uniqueness of a weakly dominant strategy equilibrium)

If the game $\Gamma = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$ has a weakly dominant strategy equilibrium s^*, then there is a non-zero probability that other weakly dominant strategy equilibria exist.

4. **Iterative elimination**

(we need for the players to be rational, and common knowledge that they are)

Let's go back to this introductory example...

Consider the following normal form game, with the following matrix representation:

		player 2		
		L	C	R
player 1	U	$4, 3$	$5, 1$	$6, 2$
	M	$2, 1$	$8, 4$	$3, 6$
	D	$3, 0$	$9, 6$	$2, 8$

And, we will combine the fact that:

- players are **rational** (hence, abiding by strict dominance)
- there is **common knowledge** that all players are rational (hence, they know that the other player will abide by strict dominance)

Are they strictly dominant strategies?

Consider the following normal form game, with the following matrix representation:

		player 2		
		L	C	R
player 1	U	$4, 3$	$5, 1$	$6, 2$
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Do player 1 and player 2 have strictly dominant strategies?

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	M	$2,1$	$8,4$	$3,6$
	D	$3,0$	$9,6$	$2,8$

Do player 1 and player 2 have strictly dominant strategies?

- **NO** for player 1
- **NO** for player 2

Are they strictly dominated strategies?

Consider the following normal form game, with the following matrix representation:

		player 2		
		L	C	R
player 1	U	$4, 3$	$5, 1$	$6, 2$
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	D	$3, 0$	$9, 6$	$2, 8$

Do player 1 and player 2 have strictly dominated strategies?

- **NO** for player 1
- **YES** for player 2, as C is strictly dominated by R

Then, **player 2 will not play C, and player 1 knows that player 2 will not play C!**

Reduce the game and do it again...

We can remove strategy C for player 2 and consider the reduced game:

		player 2	
		L	R
player 1	U	$4, 3$	$6, 2$
	M	$2, 1$	$3, 6$
	D	$3, 0$	$2, 8$

Do player 1 and player 2 have strictly dominated strategies?

Reduce the game and do it again...

We can remove strategy C for player 2 and consider the reduced game:

		player 2	
		L	R
player 1	U	$4, 3$	$6, 2$
	M	$2, 1$	$3, 6$
	D	$3, 0$	$2, 8$

Do player 1 and player 2 have strictly dominated strategies?

- **NO** for player 2
- **YES** for player 1, as both M and D are strictly dominated by U

Then, **player 1 will not play M and D, and player 3 knows that player 1 will not play M and D!**

Again and again...

We can remove strategy M and D for player 2 and consider the reduced game:

		player 2	
		L	R
player 1	U	$4, 3$	$6, 2$

Do player 1 and player 2 have strictly dominated strategies?

Again and again...

We can remove strategy M and D for player 2 and consider the reduced game:

		player 2	
		L	R
player 1	U	$4, 3$	$6, 2$

Do player 1 and player 2 have strictly dominated strategies?

- **NO** for player 1
- **YES** for player 2, as R is strictly dominated by L

Then, **we have our solution!**: based on rationality and common knowledge, the solution of that game should be (U,L), with pay-offs of 4 and 3 for players 1 and 2, respectively

Iterative Elimination of Strictly Dominated Strategies (IESDS)

The approach we have used is referred to as **Iterative Elimination of Strictly Dominated Strategies (IESDS)**, i.e.

Step 1 : Define $S_i^0 = S_i$ for player i, $i = 1, \ldots, n$. Set the counter $k = 0$

Step 2 : Are there any strategy $s_i \in S_i^k$ $(i = 1, \ldots, n)$ that are strictly dominated?

- if *yes*, go to Step 3
- if *no*, go to Step 4

Step 3 : Remove all strictly dominated strategies identified at Step 2, and for all players $i = 1, \ldots, n$. Set $k = k + 1$. Reduce the game with new strategy sets S_i^k $(i = 1, \ldots, n)$ that do not include the strictly dominated strategies. Go to Step 2.

Step 4 : The remaining strategies S_i^k $(i = 1, \ldots, n)$ are the solution, and a predictor of outcome (based on assumption of rationality and common knowledge).

And this leads to **another equilibrium concept**:

Definition (Iterated-elimination equilibrium)

Any strategy profile $s^* = (s_1^*, \ldots, s_n^*)$ that remains at the end of the IESDS process is called an *iterated-elimination equilibrium*.

Wrapping up

- **Dominance** is a key concept in analysing games, and especially normal form games
- Combined with the concepts of **rationality** and **common knowledge**, they allow to find solutions to certain games
- If a strict dominant strategy equilibrium **exists**, it is **unique**(!)
- It is not the case for both weakly dominant strategy equilibria and iterated-elimination equlibria

Further readings (/lecture notes for this topic):

Tadelis S (2013) *Game Theory - An Introduction* - Chap 4 (available with this lecture on Blackboard Learn)

There, read also the part about beliefs and best response!

Thanks for your attention!